

Pre-Judgment Legal Outcome Prediction

Graph Neural Network Architecture & Training

Dataset: Food Law Cases · 2,105 raw cases · 1,466 binary labels

Model: HGT (3-layer Heterogeneous Graph Transformer) · hidden_dim=128

0.709

Test Macro F1

0.844

Val Macro F1

76.8%

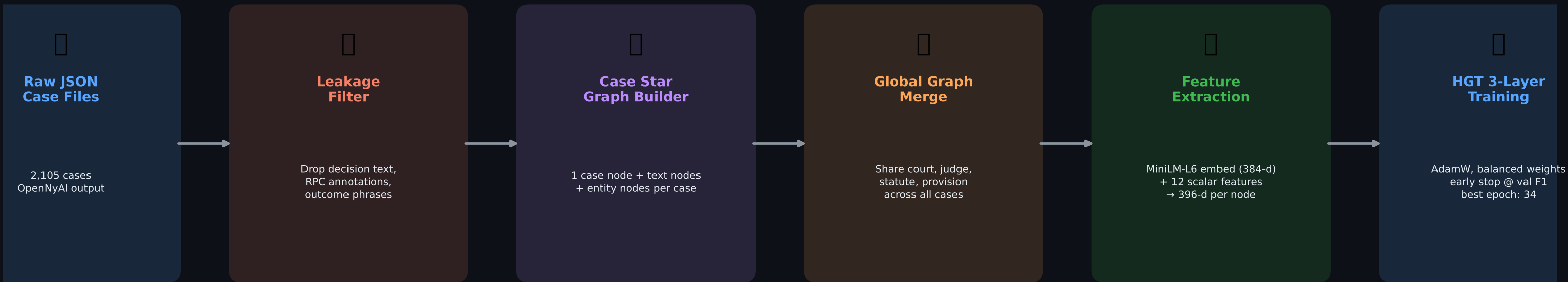
Test Accuracy

0.815

Test ROC-AUC

● End-to-End Pipeline

From raw JSON case files to binary prediction



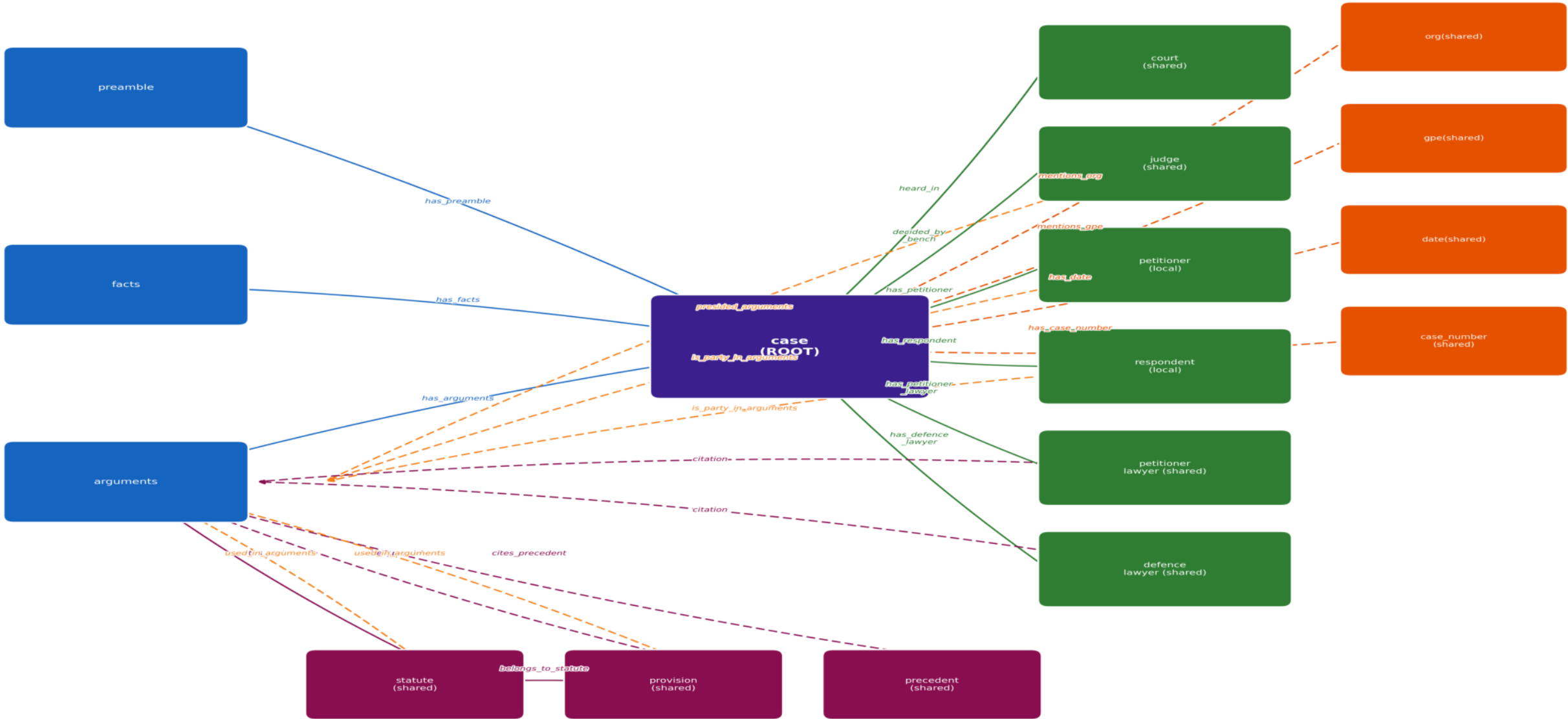
Transductive setting: full graph is always in memory; only supervision masks change.

● Graph Schema

All 18 node types and 25+ edge types in the heterogeneous graph

GNN Graph Schema — All Node & Edge Types

- Root (case)
- Text nodes (case-local)
- Entity nodes (shared)
- Legal citation nodes (shared)
- Context nodes (shared)
- Bridging / shortcut edges
- Lawyer citation edges



● Node Types

Three categories: Root · Text · Entity/Authority

ROOT

TEXT NODES
(case-local)

ENTITY / AUTHORITY



★ Shared nodes: same node object appears across every case that mentions that entity. This creates cross-case information paths.

● Edge Types & Their Role in HGT

Each (src_type, relation, dst_type) triple gets its own learned K/Q/V projections

Primary (case ↔ everything)

· case → has_preamble/facts/arguments	Case absorbs all text content
· case → heard_in → court	Which court adjudicated
· case → decided_by_bench → judge	Bench identity & history
· case → has_petitioner/respondent	Filing/opposing party
· case → has_petitioner_lawyer /	Advocate identity; shared nodes accumulate
·	has_defence_lawyer caseload signal

Citation (arguments ↔ legal entities)

· arguments → cites_statute	Law invoked
· arguments → cites_provision	Specific section invoked
· arguments → cites_precedent	Precedent relied upon
· provision → belongs_to_statute	Legal hierarchy
· petitioner_lawyer → citation → arguments	Lawyer tied to specific arguments
· defence_lawyer	→ citation → arguments Lawyer tied to defence arguments

Bridging (shortcuts to reduce hop distance)

· provision → used_in_arguments	3-hop statute→args→case becomes 2-hop
· statute	→ used_in_arguments Same shortcut for statute

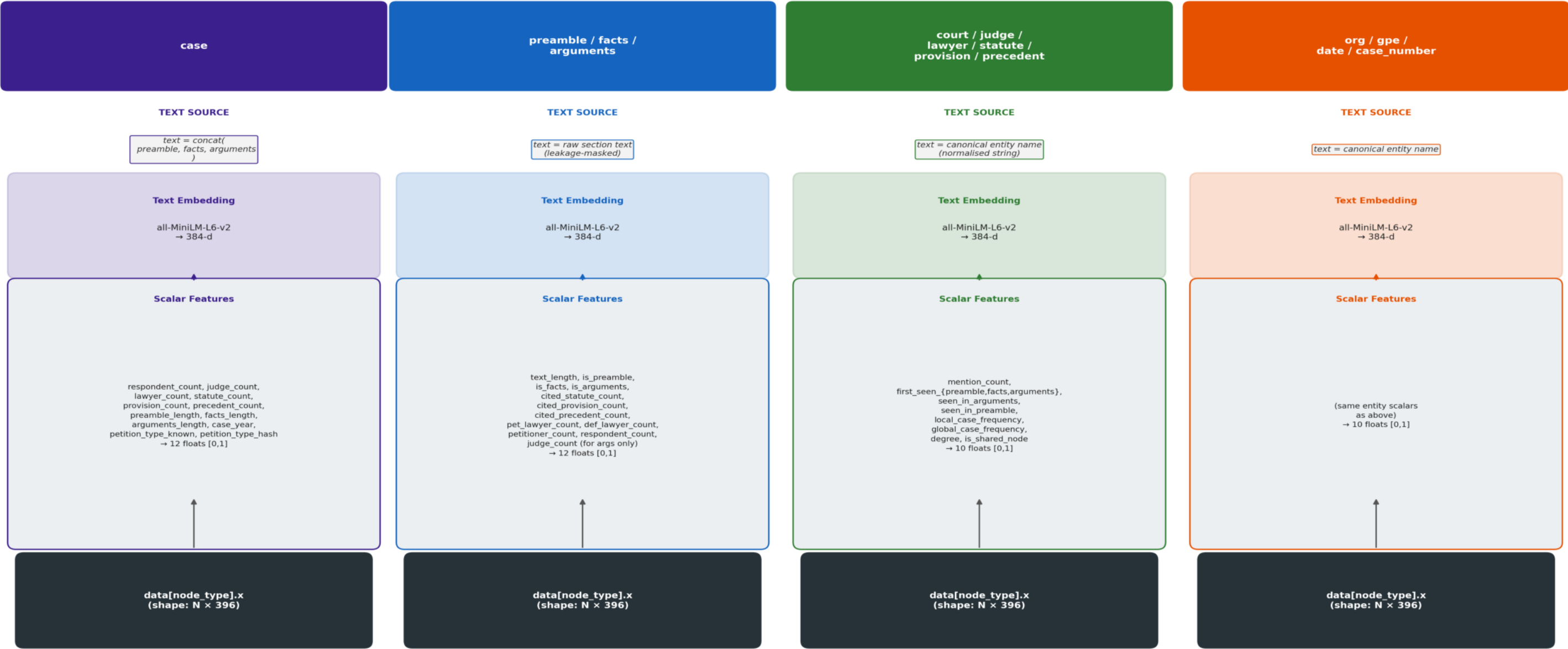
After graph construction: ToUndirected() is applied

Every directed edge gets a reverse counterpart added automatically. Messages flow symmetrically during HGT convolution.

● Node Feature Tensors — data[node_type].x

Shape: (N_nodes, 396) = 384-d text embedding + 12-d scalar features

Node Feature Storage — data[node_type].x shape: (N_nodes, embedding_dim + scalar_dim)



CONCAT(384-d embed || 12-d scalars)

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* For the fallback hashing encoder, embedding_dim = 384. * Global nodes carry global_case_frequency & degree populated during graph merge.

Layer-by-Layer Tensor State

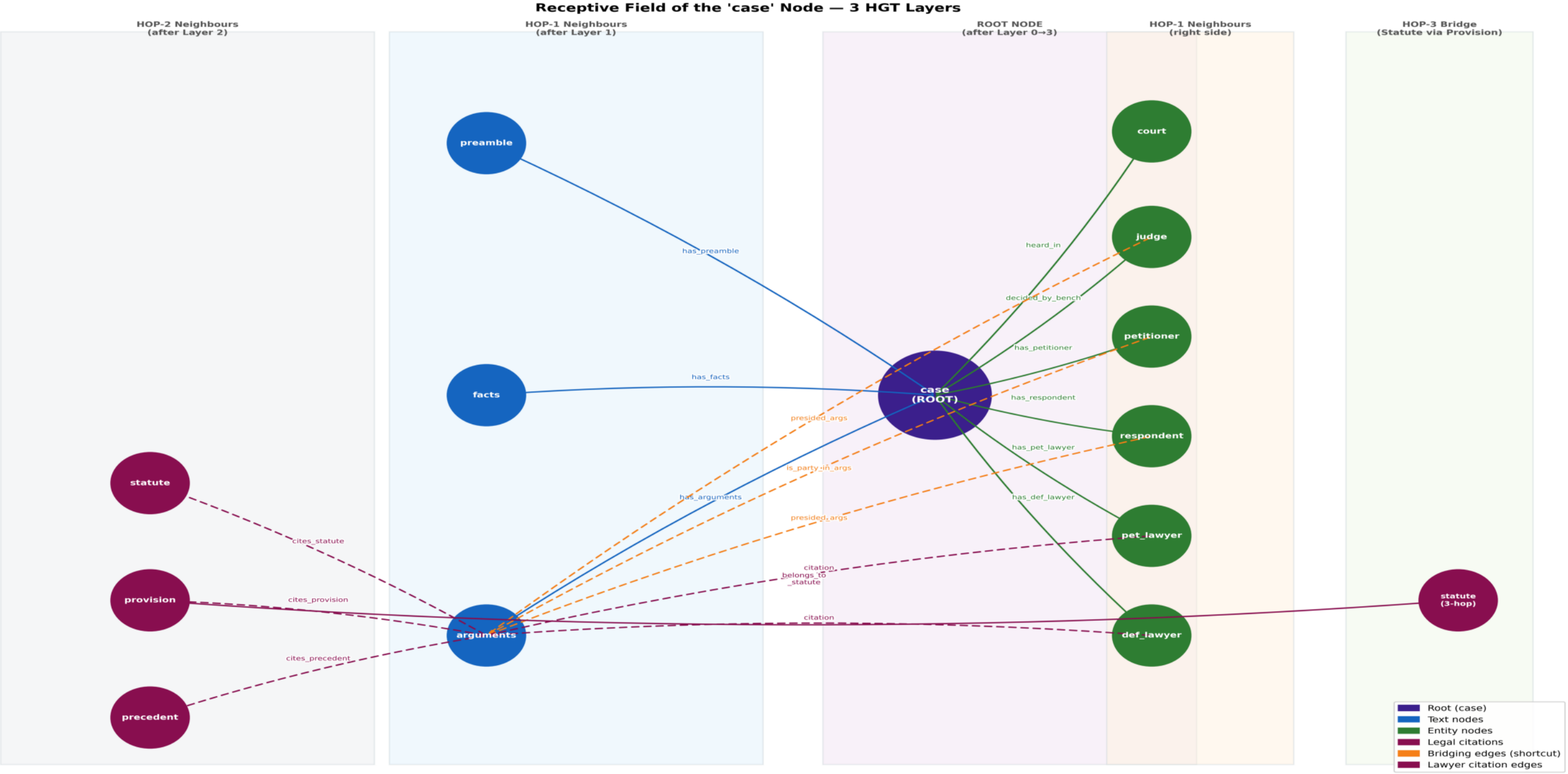
3 HGT layers: what the 'case' node absorbs at each hop

HGT 3-Layer Forward Pass — Tensor State at Each Stage



Multi-Hop Receptive Field

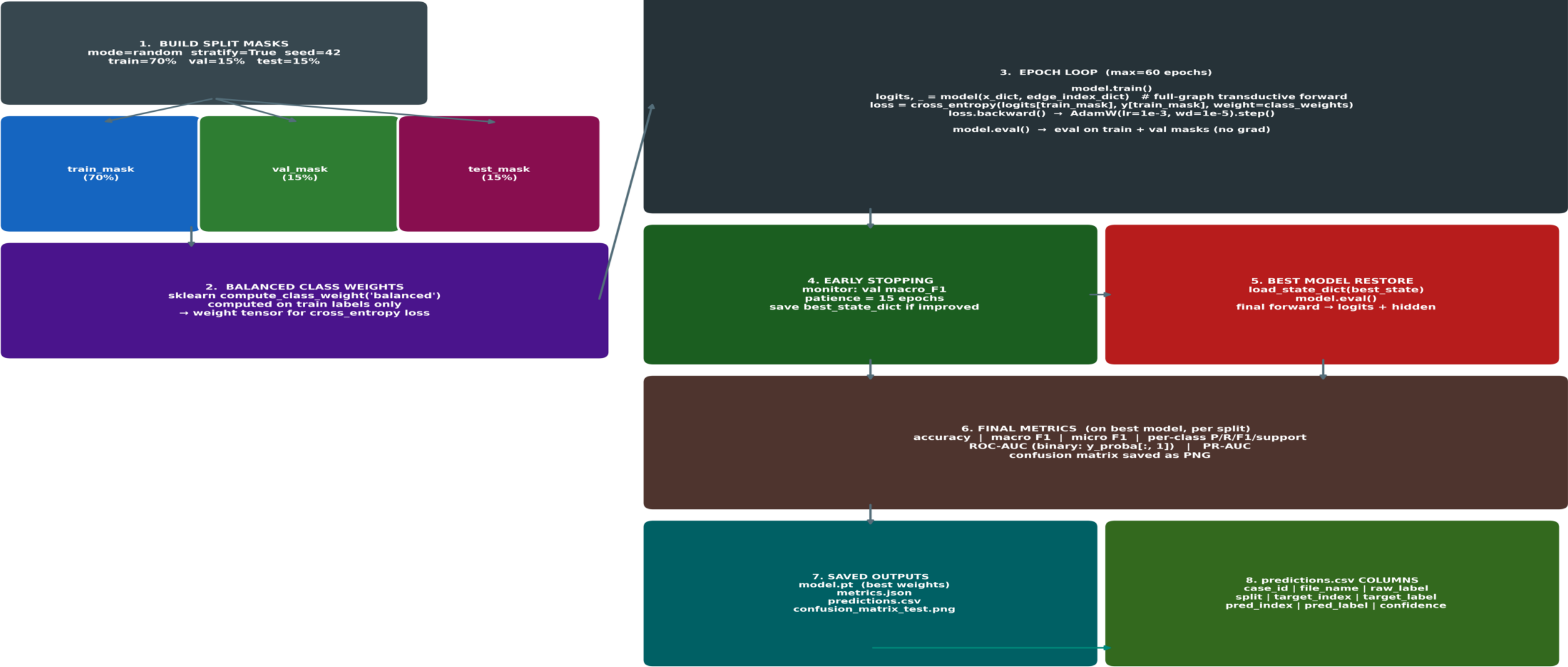
Every path the 'case' node can see after 3 HGT layers



● Training Procedure

Transductive HGT · balanced class weights · early stopping on val macro F1

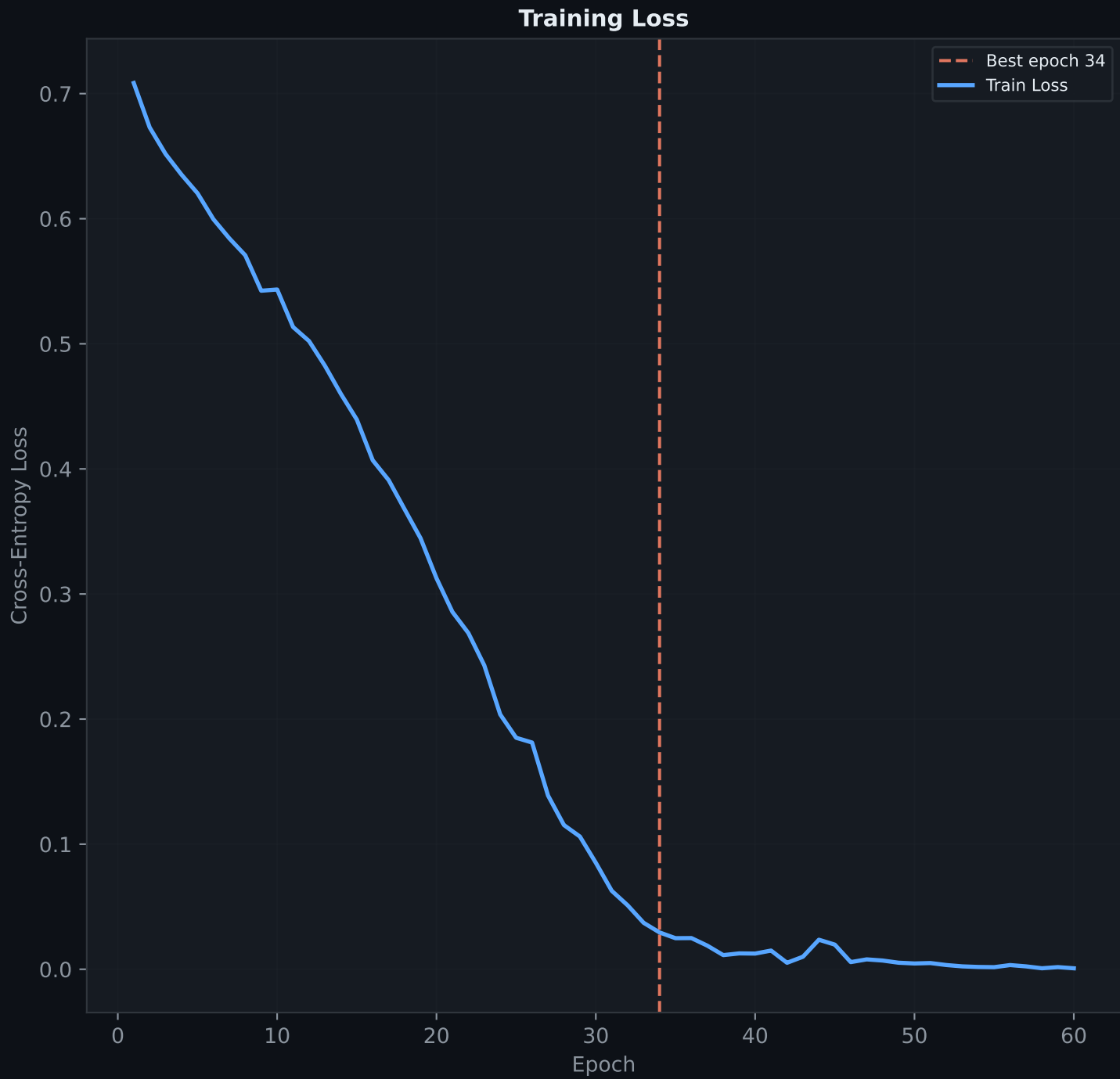
Training Loop, Early Stopping & Evaluation Procedure



★ **TRANSDUCTIVE SETTING:** HGT runs a single forward pass over the FULL graph every epoch. Masks select which case-node logits contribute to loss (train) and metrics (val/test).

● Training Curves — Seed 42 (food_law_final_full)

Best epoch: 34 — Best val macro F1: 0.8437



● Results — Best Run (Seed 42)

Binary classification: lose (0) · win (1) · Food Law dataset, 1,466 cases

TRAIN

0.994

Accuracy

0.993

Macro F1

1.000

ROC-AUC

1.000

PR-AUC

VAL

0.868

Accuracy

0.844

Macro F1

0.896

ROC-AUC

0.948

PR-AUC

TEST

0.768

Accuracy

0.709

Macro F1

0.815

ROC-AUC

0.888

PR-AUC

Per-Class Results (Test Set — Seed 42)

Class	Precision	Recall	F1	Support
lose	0.673	0.507	0.579	69
win	0.798	0.887	0.840	151

● Limitations & Improvement Directions

What currently constrains performance and how to address it

IMPROVEMENT DIRECTIONS

⚠️ Class Imbalance

win:lose $\approx$ 2.2:1 in binary set.
Balanced class weights help but 'lose' F1 still lags (0.578 test).

⚠️ Overfitting

Train F1 $\rightarrow$ 1.00 while test F1 = 0.709.
Gap suggests model memorises training cases.

⚠️ Transductive Leakage Risk

Even with masks, ALL nodes (incl. test) participate in message passing.
Shared authority nodes carry indirect test-case signal.

⚠️ String-based Entity Normalisation

Merging authority nodes relies on fuzzy string matching.
Noise in entity resolution can mis-merge or fragment nodes.

✦ Mini-batch HGTLoader

Inductive training; removes transductive leakage.

✦ Chunked Text Embeddings

Mean-pool 512-token chunks instead of truncating long arguments.

✦ 3-Seed Averaging

Runs 2 & 3 in progress; average final test metrics.

✦ Adversarial Regularisation

Add noise to shared node embeddings at train time.

✦ Temporal Split

Use year-based split to stress-test out-of-distribution generalisation.